

MQTT + ESP32 + WS2812B — Student Reference Card

Camp Day 2 | Keep this sheet handy!

1 Wiring (breadboard)

ESP32 Pin	LED Strip Wire	Notes
GND	GND (black)	Must be shared!
5V / VIN	+5V (red)	Power for LEDs
GPIO 5	DIN / IN (green)	330 ohm resistor in series recommended

IMPORTANT: Connect to DIN (IN) end of strip, NOT DOUT.

2 Thonny Setup (do once)

1. Plug ESP32 in with a **data USB cable** (not charge-only).
2. Open Thonny -> Tools -> Options -> Interpreter.
3. Choose **MicroPython (ESP32)**.
4. Select your port: Windows = COM3/COM4 Mac = /dev/cu.usbserial-... Linux = /dev/ttyUSB0
5. Click OK. Shell should show >>>
6. Open **main.py**, fill in your Wi-Fi and Pi IP, then press the green Run button.

3 Send Color Commands (from Raspberry Pi terminal)

Command (type on the Pi)	What it does
<code>mosquitto_pub -h localhost -t led/cmd -m "red"</code>	All LEDs -> Red
<code>mosquitto_pub -h localhost -t led/cmd -m "green"</code>	All LEDs -> Green
<code>mosquitto_pub -h localhost -t led/cmd -m "blue"</code>	All LEDs -> Blue
<code>mosquitto_pub -h localhost -t led/cmd -m "white"</code>	All LEDs -> White
<code>mosquitto_pub -h localhost -t led/cmd -m "off"</code>	All LEDs -> Off

4 Quick Troubleshooting

- **No >>> in Thonny shell:** Try a different USB cable or USB port. Wrong cable is the #1 problem.
- **LEDs don't light at all:** Check DIN vs DOUT. Confirm LED_PIN = 5 matches your wiring.
- **ESP32 resets when LEDs connect:** Power issue. Keep BRIGHTNESS low. Ask instructor for a 5V supply.
- **MQTTException / no color change:** Check MQTT_BROKER IP. ESP32 and Pi must be on same Wi-Fi network.
- **ImportError: no module 'umqtt':** Ask instructor for the umqtt folder to copy onto the ESP32.
- **LEDs blink dim red slowly:** Wi-Fi failed. Double-check SSID and password in main.py.
- **LEDs solid dim blue:** Good! Connected and ready. Send a color command from the Pi.

5 'Is It Working?' Checklist

- [] Pi broker test: one terminal subscribes, another publishes hello — you see hello appear.
- [] ESP32 starts -> LEDs glow **dim blue** (ready indicator).
- [] From Pi: `mosquitto_pub ... -m "red"` -> LEDs turn red.

Read troubleshooting above carefully before asking for help — most problems are listed there.